



Review

Hospital Care for Pregnant Persons With OUD Review[☆]Joan Park, MD^{a,*}, Mishka Terplan, MD^b, Jessica Ratner, MD^a^a Division of Addiction Medicine, Johns Hopkins School of Medicine, Baltimore, Md^b Friends Research Institute, Baltimore, Md

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ABSTRACT

Opioid use disorder is increasingly prevalent among pregnant persons. Hospitalists and general obstetricians are uniquely positioned to treat pregnant persons with OUD, who may not otherwise be engaged in care. Pregnant people with OUD face stigma, discrimination, and fear of losing custody of their children. Providers should offer supportive and nonjudgmental care that is guided by the principles of harm reduction. All pregnant persons should be screened for OUD, diagnosed and treated accordingly. Medications for opioid use disorder (MOUD), methadone and buprenorphine, are safe in pregnancy and are gold standard treatment. Patients should be initiated or continued on MOUD during hospitalization. Pregnancy may increase metabolism of both buprenorphine and methadone, and therefore, dosages may need to be increased or given more frequently. Given the increase of high potency synthetic opioids, such as fentanyl, providers should be familiar with low-dose buprenorphine initiation or rapid methadone titration. Providers should monitor for opioid withdrawal and treat aggressively with MOUD and adjuvant medications. This may also require full agonist opioids. OUD treatment and support should continue into the post-partum period, a time of increased vulnerabilities to return to use, OUD recurrence, and overdose. Discharge planning should be family-centered, considering partners, their family, and the newborn. Social work and peer navigators should be engaged to help coordinate long-term outpatient resources including postpartum psychosocial support services such as substance use disorder treatment and relapse prevention programs.

Introduction

Opioid use disorder (OUD) is increasingly prevalent among pregnant persons. Maternal opioid use disorder diagnoses increased from 3.5 to 8.2 per 1000 delivery hospitalizations from 2010 to 2017,¹ and by 2019, mental health conditions, which include overdose, were the leading cause of pregnancy-related deaths nationally.² This risk is further heightened by the rise of fentanyl and prevalence of polysubstance use among pregnant people with OUD.^{3,4} Beyond overdose, recent studies demonstrate that pregnant persons with OUD have double the odds of experiencing significant maternal morbidity, including cardiac arrest, sepsis, placental abruption, preterm birth.⁵ Yet only 34.7% of pregnant persons with OUD report receipt of treatment.⁶ The prenatal period is an opportunity for care engagement as pregnancy can be a strong motivator to seek treatment.⁷

Hospitalists and general obstetricians are uniquely positioned to initiate treatment and assist with transitions to ongoing care in the outpatient setting for pregnant persons with OUD. Pregnant persons are hospitalized for birth and sometimes for other acute medical needs. These admissions provide a unique opportunity to engage patients who may

not be engaged in primary care or addiction treatment and connect them to longitudinal care.⁸ Moreover, contrary to the general population, in the pregnant population, MOUD titration is an appropriate and reimbursable condition for inpatient admission. Studies in the general population have shown that hospital-based addiction care including both consultation and peer navigation services can improve addiction treatment engagement and reduce substance use.^{9,10}

This review seeks to provide guidance for practitioners on the non-obstetric management and in-hospital treatment of opioid use disorder in pregnant persons. At current, there is no guideline for in-patient providers on how to care for pregnant persons with OUD in the hospital. To conduct this narrative review, we searched PubMed in January 2025 using terms including “opioid use disorder,” “pregnancy,” and “hospitalizations.” Results were reviewed for relevant citations. Additional references were identified through the review of these articles.

General Approach

When taking care of pregnant persons with opioid use disorder, providers should offer supportive and nonjudgmental care. Pregnant

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Table 1
Harm Reduction Examples in the Hospital

Situation	Harm Reduction Response
Patient reports recent drug use on admission	• Explore potential withdrawal symptoms that may affect in-hospital care and patient interest in treatment after discharge • Do not strip search • Offer opportunity to safely store belongings during admission • Provide naloxone on discharge
In-hospital drug use	• Do not reflexively administratively discharge • Request that patient not use substances in hospital • Offer opportunity to dispose of substances or lock belongings in a secure place • Explore with patient reasons for use and address where able (e.g., untreated withdrawal or cravings) • Limit involvement of security or law enforcement to imminent risk of harm for patient, staff, or others
Patient not interested in OUD treatment after discharge	• Support patient autonomy, explore motivations, and build rapport • Use shared decision-making to identify preferred method of treating withdrawal while admitted • For prolonged admissions, consider titrating to a therapeutic dose of methadone or buprenorphine to preserve tolerance and reduce risk of post-discharge overdose
Patient overdose in hospital	• Link patients to harm reduction services • Make naloxone easily accessible on every unit • Consider monitoring with pulse oximetry and reserving naloxone for clinically significant respiratory depression rather than sedation alone (to minimize precipitated withdrawal and patient discomfort) • Evaluate for iatrogenic causes or contributions (e.g. medications, medical conditions)
Patient desires self-directed discharge	• Try to quickly mobilize discharge resources, including: ○ Linkage to outpatient follow-up, including medications for OUD ○ Prescription of oral antibiotics, when needed ○ Access to community harm reduction services ○ Dispensing (rather than prescribing) naloxone

people with OUD face stigma, discrimination, and fear of losing custody of their children, which they report are barriers to treatment.⁷ Consistency and clear, open communication are important to maximize engagement. Care should be guided by the principles of harm reduction, which aims to decrease negative impacts of substance use, rather than focusing exclusively on abstinence. Harm reduction seeks to respect patient autonomy through creating a safe environment for care, allowing patients to guide their care based on their goals, and providing the information needed for the patient to make informed decisions for their care.¹¹ This includes respecting the patient’s right of refusal for testing and treatment. [Table 1](#) offers potential ways that a harm reduction approach can be applied to challenging situations that may arise in the hospital.

Care for the pregnant person should involve a multidisciplinary team, ideally including a prenatal care provider, pediatrician, addiction specialist, peer navigators, nurses, and social workers, when available. Peer navigators can offer significant benefits for care coordination, patient understanding, and access to health care and social service systems. During the perinatal period, peer navigator engagement is associated with decreased substance use and improved maternal and neonatal health outcomes.¹²

INITIAL ASSESSMENT

Screening for Substance Use

All pregnant persons should be screened for substance use using validated verbal or written instrument, and those who screen positive should be evaluated further for diagnosis and appropriate treatment. Polysubstance use is common amongst pregnant persons and may impact how OUD presents.⁴ Patients may be reticent to disclose substance use due to fear it will lead to child welfare engagement and stigma.⁵ Therefore, providers should communicate the purpose of screening - to provide treatment and support – and obtain informed consent.¹³ ACOG recommends the following validated screening tools: 4Ps, NIDA Quick Screen, and CRAFFT.¹³

Providers should avoid targeted screening but rather screen all pregnant persons. Multiple studies have found that when providers target their assessment, non-Hispanic Black and Hispanic patients are more likely to be tested compared to White patients.^{14,15} This propagates stigma amongst certain populations while missing SUD in others.

Providers should evaluate patients who screen positive for a use disorder using Diagnostic and Statistical Manual of Mental Disorders (DSM) criteria (see below) and treat as indicated. Both patients with risky substance use and a use disorder can benefit from motivational interviewing to explore the patient’s goals while pregnant, educate on the impact of opioid use during pregnancy, and offer harm reduction strategies.

Use of Urine Toxicology

Urine toxicology should not be used as a screening tool for opioid use disorder. Rather, use of toxicology testing can be considered when it might help guide clinical care– for example, to guide harm reduction counseling around fentanyl overdose and concurrent benzodiazepine use.¹⁶ Professional society recommendations are clear: informed consent is required, and patients have the right to refuse testing.¹³ They should be informed on the purpose of toxicology testing and the limits around confidentiality based on state-specific mandatory reporting laws. Providers can refer to the If/When/How resource on Prenatal Drug Exposure for more details.¹⁷

It is important to note that a positive test does not prove an opioid use disorder, nor does a negative test rule out substance use. Urine toxicology test interpretation can be complicated, and providers should be familiar with its limitations. False positives are common in immunoassay testing (the most common type used for initial testing) because of cross-reactivity with substances that have similar structures to illicit drugs.^{18,19} False negatives can also occur when the concentration of the drug is less than the assay cut off.²⁰ Additionally, true positives may sometimes reflect medications administered during the course of clinical care; for example, a recent study demonstrated that neuraxial fentanyl, a common method for labor analgesia, led to positive maternal and neonatal toxicology in 80% of both mothers and babies.²¹ Any unexpected result should be discussed with the patient, and a confirmatory mass spectrometry test should be done as a follow-up, if toxicology results conflict with self-reported use.

Diagnosis

Opioid use disorder is diagnosed based on DSM-V criteria. The eleven criteria fall into four categories: 1) impaired control (using more than intended; being unable to cut down); 2) social problems (neglecting responsibilities; continued use despite interpersonal problems exacerbated by opioids; sacrificing important activities); 3) risky use (using in risky settings; continued use despite known problems), and 4) physical dependence (tolerance and withdrawal). Severity is determined by the number of criteria the patient meets: 2-3 symptoms indicate a mild use disorder; 4-5 symptoms indicate moderate; 6 or more indicates severe. Of note, when opioids are prescribed (e.g., for chronic pain), tolerance

and withdrawal alone in the absence of other positive DSM-V criteria do not indicate the presence of an opioid use disorder.

Screening Beyond SUD

All pregnant persons diagnosed with opioid use disorder should be screened for sexually transmitted infections, including HIV, chlamydia, gonorrhea, syphilis, and other infections, such as hepatitis B, hepatitis C, and tuberculosis, and offered treatment when appropriate. Patients who use substances are at higher risk of co-occurring mental health disorders and intimate partner violence and should be screened for both.^{22,23} Providers should also screen for social determinants of health (i.e. food insecurity, housing instability, IPV) which can and should be intervened on during hospitalization.

Clinical Management of Opioid Use Disorder

Medication for Opioid Use Disorder (MOUD)

The treatment of opioid use disorder in pregnancy is similar to that of the general population. Methadone and buprenorphine are the standard of care for opioid use disorder. They are the safest and most effective medications for OUD in pregnancy, decreasing rates of overdose, return to use, and infection.²⁴ A 2020 Cochrane review concluded that buprenorphine and methadone have similar efficacy for treatment of OUD during pregnancy.²⁵ If a pregnant person is taking methadone or buprenorphine prior to pregnancy and finds it effective, it should be continued. If the patient is not on MOUD, providers should initiate methadone or buprenorphine based on patient preference, potential drug interactions, and follow-up considerations (e.g. Can patient go to methadone clinic on a daily basis? Would patient benefit from daily dosing or supervision?). MOUD should be started as soon as possible—A large retrospective cohort study demonstrated that longer duration of treatment with MOUD during pregnancy was associated with improved perinatal health outcomes and decreased rates of maternal overdose.²⁴

MOUD during pregnancy is superior to medically-assisted withdrawal (also known as “detox”). Patients may desire medically-assisted withdrawal in hopes of preventing Neonatal Opioid Withdrawal Syndrome (NOWS). However, this rarely prevents NOWS due to low completion rates and high rates of return to use afterwards.²⁶ Medically assisted withdrawal is an acute intervention that does not address addiction’s chronic and relapsing cycles. If patients would like medically-assisted withdrawal, providers should explore their reasons (e.g., family or partner desires, fear of court involvement, desire to prevent NOWS), offer education to correct any misinformation, and use shared decision-making to determine the best path forward.

Safety in Pregnancy

Methadone and buprenorphine are safe in pregnancy. Both medications provide more stable levels of opioid blood concentrations than illicit opioids, which reduces stress on the fetus. Several large cohort studies have investigated whether buprenorphine or methadone are associated with congenital anomalies. Overall the literature is reassuring; rates are low and inconsistent between studies, and a biological pathway for birth defects has not been detailed.²⁷⁻²⁹ Hence, both buprenorphine and methadone are considered safe and effective in pregnancy.

Methadone and buprenorphine can result in physiologic dependence in utero leading to NOWS. Though absolute rates are similar between the two medications, buprenorphine is associated with decreased severity and duration of withdrawal compared to methadone.³⁰ Neither methadone nor buprenorphine dosages have been consistently found to correlate with the development or severity of NOWS.³¹ Providers should counsel patients that NOWS is a treatable and time limited outcome of maternal opioid exposure, while MOUD is lifesaving. Thus, patients

Table 2

Standard Dose Initiation for Buprenorphine Example.

Start	2 mg sublingual buprenorphine tablet/film for 1 dose
1 hour later	2 mg sublingual buprenorphine tablet/film for 1 dose (if previous dose tolerated)
3 hour later	4 mg sublingual buprenorphine tablet/film for 1 dose (if previous dose tolerated)
4 hour later	8 mg sublingual buprenorphine tablet/film twice a day

Institutional practice based on ASAM National Practice Guideline for the Use of Medication in Treatment of Addiction Involving Opioid Use, Special Populations: Pregnant Women.⁴¹

should be treated with the MOUD type and dose that most effectively treats their cravings and withdrawal. Concomitant benzodiazepines, gabapentin, tobacco, and other psychotropic medications may influence the severity and duration of NOWS.^{32,33,34} If there are concerns about co-exposures, a perinatal psychiatrist should be consulted rather than tapering medications and risking mood destabilization. Providers can seek psychiatric consultation by calling Postpartum Support International (PSI) Perinatal Psychiatric Consult, a national perinatal mental health support line. Many states also provide provider-to-provider psychiatric consultations as well. The National Maternal Mental Health Hotline is available 24/7 for patients experiencing mental health concerns during pregnancy or the postpartum period, offering free, confidential support and referrals to local resources.

Buprenorphine

Buprenorphine is a partial agonist with a high affinity for the opioid receptor. It is available as a monoproduct (e.g., Subutex) or in a combined formulation with naloxone (e.g., Suboxone or Zubsolv). Naloxone has minimal bioavailability in sublingual formulations; its presence in the combined formulation is designed to deter abuse, as injection can potentially cause precipitated withdrawal. The monoproduct has historically been recommended over the combined product during pregnancy given lack of safety data on naloxone during pregnancy. However, recent research suggests that both formulations are equally safe, and shared decision-making can be used to guide formulation choice.³⁰ Buprenorphine also comes as a long-acting injectable (e.g., Brixadi and Sublocade). Evidence for its use in pregnant persons is still emerging and its uptake in the inpatient setting is limited at this time.

Patients who are well maintained on buprenorphine should be encouraged to continue through pregnancy. Providers should use the Prescription Drug Monitoring Program (PDMP), a state-run database that tracks controlled substance prescriptions, to confirm the most recent prescription. Buprenorphine is metabolized by the CYP3A4 pathway, which is increased during pregnancy. Therefore, a dose frequency increase may be necessary to manage patients’ withdrawal and cravings (e.g., dosing every 6-8 hours).³⁵ If a switch to methadone is required, buprenorphine can be stopped and methadone initiated, as outlined below. The addition of a full mu-opioid agonist (e.g., methadone) to a partial agonist (e.g., buprenorphine) does not lead to a drug interaction that causes withdrawal.

As a partial agonist with high affinity for the mu opioid receptor, buprenorphine can precipitate withdrawal if initiated while receptors are saturated with full agonist opioids (e.g., fentanyl, heroin, oxycodone). Two potential approaches to buprenorphine initiation – “standard dose initiation” (Table 2) and “low dose initiation” (Table 3) are outlined below.

To initiate buprenorphine using standard dose initiation, patients must wait for clear signs of opioid withdrawal indicating sufficient opioid receptor availability prior to their first dose. Symptoms should be monitored using a standardized instrument, such as the Clinical Opioid Withdrawal Scale (COWS, see Table 4). Once signs of mild-moderate opioid withdrawal (COWS > 8) are present, the patient can start the

Table 3

Low Dose Initiation for Buprenorphine Example.

Start	Buccal Buprenorphine 150 mcg every 6 hours for 2 doses
6 hours later	Buccal Buprenorphine 300 mcg every 6 hours for 2 doses
6 hours later	Sublingual Buprenorphine 2 mg every 6 hours for 2 doses
6 hours later	Sublingual Buprenorphine 4 mg every 6 hours for 2 doses
6 hours later	Sublingual Buprenorphine 8 mg twice a day

Prior to initiation, patient should be started/continued on full agonists opioids. Only after patient is comfortable and withdrawal symptoms addressed, should buprenorphine be initiated. After patient received two doses of buprenorphine 8 mg, full agonists can be stopped.

The maximum total daily dose of buprenorphine is 32 mg.

Institutional practice recommendations. Current evidence is primarily case-based, in particular within the prenatal population.⁴² There is no available data to recommend a specific dosing schedule.⁴³ For hospitals that do not have buccal buprenorphine available, equivalent doses of intravenous or intramuscular buprenorphine ("Buprenex") can be used instead. There are some institutions that have used transdermal buprenorphine for initiation as well.

standard dose initiation (Table 2). Dose can be titrated based on cravings and withdrawal symptoms. Studies have consistently shown that dosages greater than 16 mg and as high as 32 mg are associated with improved outcomes.^{36,37,38}

Given methadone's long-half life, providers should be cautious when transitioning patients from methadone to buprenorphine using standard dose initiation and consider low-dose initiation, as below.

In recent years, fentanyl has become prominent amongst the illicit drug supply.³⁹ Patients are at higher risk of precipitated withdrawal due to fentanyl's lipophilic nature.⁴⁰ Patients may be hesitant to initiate buprenorphine using the standard approach either due to difficulty tolerating withdrawal with opioid cessation or fear of precipitating withdrawal with buprenorphine. A low dose initiation strategy may be helpful for these patients. This strategy utilizes very small initial dosages of buprenorphine with incremental increases (Table 3). Prior to initiation, patients should be comfortable without withdrawal symptoms. This often requires that they be started on either low dose methadone or scheduled short acting agonists. Buprenorphine will accumulate at the mu-receptor and gradually replace the full agonist opioid. When buprenorphine has reached 8 mg twice a day, full agonist opioids may be discontinued.

While this method is often well tolerated, there is a small chance of precipitated withdrawal. In this case, next steps should be determined based on patient preference. If the patient wants to continue with initiation, they can return to the prior tolerated dose and continue that dose until they are comfortable before proceeding with the up-titration. An alternative is to give 2 mg buprenorphine every hour until symptoms abate. If the patient wishes to stop the initiation, buprenor-

Table 4

Clinical Opiate Withdrawal Scale

Resting Pulse Rate (BPM)	0 ≤80
Measure pulse rate after patient is sitting or lying down for 1 minute	1 81-100
	2 101-120
	4 >120
Sweating	0 No reports of chills or flushing
Sweating not accounted for by room temperature or patient activity over the last 0.5 hours	1 Subjective report of chills or flushing
	2 Flushed or observable moistness on face
	3 Beads of sweat on brow or face
	4 Sweat streaming off face
Restlessness observation during assessment	0 Able to sit still
	1 Reports difficulty sitting still, but is able to do so
	3 Frequent shifting or extraneous movements of legs/arms
	5 Unable to sit still for more than a few seconds
Pupil Size	0 Pupils pinned or normal size for room lights
	1 Pupils possibly larger than normal for room light
	2 Pupils moderately dilated
	5 Pupils so dilated that only the rim of the iris is visible
Bone or joint aches	0 Not present
If patient was having pain previously, only the additional component attributed to opiate withdrawal is scored	1 Mild diffuse discomfort
	2 Patient report severe diffuse aching of joints/muscles
	4 Patient is rubbing joints or muscles and is unable to sit still because of discomfort
Runny nose or tearing	0 No present
Not accounted for by cold symptoms or allergies	1 Nasal stuffiness or unusually moist eyes
	2 Nose running or tearing
	4 Nose constantly running or tears streaming down cheeks
GI upset	0 No GI symptoms
Over the last 0.5 hours	1 Stomach Cramps
	2 Nausea or loose stool
	3 Vomiting or diarrhea
	5 Multiple episodes of vomiting or diarrhea
Tremor observation of outstretched hands	0 No tremor
	1 Tremor can be felt, but not observed
	2 Slight tremor observable
	4 Gross tremor or muscle twitching
Yawning observation during assessment	0 No yawning
	1 Yawning once or twice during assessment
	2 Yawning three or more times during assessment
	4 Yawning several times/minute
Anxiety or irritability	0 None
	1 Patient reports increasing irritability or anxiousness
	2 Patient obviously irritable/anxious
	4 Patient so irritable or anxious that participation in the assessment is difficult
Gooseflesh skin	0 Skin is smooth
	3 Piloerection of skin can be felt or hairs standing up on arms
	5 Prominent piloerection

Table 5
Adjuvants for Opioid Withdrawal for Pregnant Persons.

Withdrawal Symptom	Medications
Rhinorrhea	Cetirizine, Loratadine, Fexofenadine, Diphenhydramine, Pseudoephedrine, Guaifenesin
Nausea/Vomiting	Diphenhydramine, Doxylamine/Pyroxidine, Metoclopramide, Compazine, Promethazine, Ondansetron (<i>avoid in first trimester</i>)
Heartburn	Aluminum hydroxide/Magnesium hydroxide/Simethicone, Famotidine, Omeprazole
Diarrhea	Loperamide, Kaopectate
Pain	Acetaminophen (<i>avoid NSAIDS</i>)
Insomnia	Doxylamine, Diphenhydramine (use sparingly)
Anxiety/Autonomic Symptoms	Hydroxyzine, Clonidine (use sparingly and in discussion with obstetrician), Benzodiazepines (use sparingly), Gabapentin (use sparingly)

Table 6
Rapid Methadone Titration.

Day	
1	Give methadone 30 – 40 mg. If persistent withdrawal after 4 hours, give 10 mg for up to 2 doses. *** On day 1, maximum total daily dose is 60 mg.
2	Give the previous day's total dose as new AM dose. If COWS > 9 after 4 hours, can give an additional 10 mg for up to 2 doses.
3+	Continue this process until reaching morning dose of methadone 100 mg. Hold for 3 days. Increase methadone by 10 mg every 3-4 days, as needed

Institutional practice based on retrospective cohort study.⁴⁴ There are several case reports and retrospective cohort studies with variable rapid methadone initiation protocols for pregnant individuals. There is no available data to suggest a specific protocol is superior.

phine should be stopped. They can then receive full agonist opioids to overcome withdrawal symptoms (e.g., with hydromorphone). Providers should treat withdrawal symptomatically with the adjuvants discussed below (Table 5). These principles also apply for precipitated withdrawal with standard dose initiation of buprenorphine.

Methadone

Methadone is a long-acting full agonist of the mu-opioid receptor. There is significant inter-individual variability in methadone pharmacokinetics; in particular, the half-life of methadone can vary from 8 to 59 hours depending on the patient. Because of these pharmacokinetic features, methadone initiation has historically been started a low dosage (e.g., 30mg daily) and slowly increased (e.g., by 10mg every 3-7 days). However, in the era of fentanyl, patients have had higher tolerances that require higher dosages of methadone.³⁸ In an effort to more effectively and efficiently address patients' withdrawal, there has been a push towards faster titration. Several retrospective studies have published encouraging results on the safety and efficacy of this approach.⁴⁴ These faster methadone titrations typically start with a total daily dose of methadone 30-60 on day 1 and increase by 10-20 mg every day (see Table 6 for an example). Patients who meet all of the following criteria can be considered for faster titration: 1) treated with methadone 80 mg or higher within the last year; 2) reports using fentanyl at least daily prior to admission; 3) has consistent opioid use over the last 2 weeks; 4) do not have end-organ failure including kidney or liver (given effect on methadone metabolism) or pulmonary disease (given potential respiratory compromise). Faster titrations should be avoided in patients with ventricular arrhythmias, QTc > 500 ms, or concurrent use of sedatives or other medications that may affect methadone metabolism. These high-risk patients can be titrated using the standard protocol (see Table 7 for an example). Regardless of the titration schedule utilized, uptitration should continue until a dose is reached that controls a patient's opioid

Table 7
Standard Methadone Titration.

Day		Maximum Total Daily Dose
1	Give methadone 30 mg. If persistent withdrawal after 6 hours, can give an additional 10 mg once.	40 mg
2	Give the previous day's total dose as new AM dose. If persistent withdrawal after 6 hours, can give an additional 10 mg.	50 mg
3	Continue this process until reaching morning dose of methadone 60 mg. Hold for 3 days.	60 mg
4+	Increase methadone by 10 mg every 3-4 days, as needed.	

Institutional practice based on ASAM National Practice Guideline for the Use of Medication in Treatment of Addiction Involving Opioid Use, Special Populations: Pregnant Women.⁴⁷

withdrawal symptoms and cravings for a full 24-hour period without causing side effects.

If a patient is already on methadone, it should be continued in the inpatient setting. In the outpatient setting, methadone requires administration through an opioid treatment program (OTP) that is certified by Service Abuse and Mental Health Services Administration (SAMHSA). Providers must contact the patient's OTP to confirm their dose and date of last dispense. Until the dose can be confirmed, methadone 40 mg can be given along with short-acting opioids to manage withdrawal.

As pregnancy progresses, CYP3A4 expression increases, increasing methadone's metabolism.⁴⁵ A single dose may not alleviate symptoms for 24 hours. Therefore, patients may benefit from split dosing (every 12 hours), particularly in the third trimester.⁴⁶ This decision should be made on an individual basis. Any dose changes made in the inpatient setting should be communicated with the patient's OTP. Take note that some OTP's may not be able to offer split dosing for patients in the outpatient setting.

For patients who have missed doses of methadone, reinitiation depends on the number of doses missed. There are no published guidelines on methadone re-initiation for patients with missed dosages. Our institutional approach is as follows: If one or two doses of methadone are missed, the patient can be maintained on the same methadone dose. If three doses are missed the next methadone dose should be reduced by 25% to adjust for potential reduction in tolerance. If the reduced dose is well tolerated, the following day's dose can return to previous dose levels. If four doses are missed the next dose should be reduced by 50% to adjust for reduction in tolerance. If the dose is well tolerated, they can receive 75% of their dose the following day and 100% of their dose in 2 days. If more than four doses are missed, patients should have methadone induction from baseline.

Methadone may cause sedation or respiratory depression. If possible, patients should be evaluated at peak serum levels, around 4 hours after dosing. If patients are sedated after their dose of methadone, then the next daily dose should be reduced (and the patient monitored until they are no longer sedated or treated for overdose if necessary). Patients should be observed each day prior to dosing. Patients who are sedated or intoxicated should not be given further doses of methadone until the sedation has abated.

While methadone can, in some cases, cause QT prolongation, it is very rarely the sole cause of QT prolongation, and the average magnitude of QTc change from methadone is generally modest (+10-12 ms). Conversely, disruption of methadone treatment for opioid use disorder confers an extremely high risk of death, particularly in the context of active, severe disease. QTc should be monitored at methadone initiation and when increasing the dose above 120 mg/d. If elevated, providers should address other modifiable risk factors such as avoidance of other potentially QTc-prolonging medications and correction of electrolytes (considering high K and Mg targets of 4 mg/dL and 2 mg/dL, respectively).

Pregnant people well maintained on methadone should be encouraged to continue through pregnancy. If transition to buprenorphine is required, studies suggest it is possible to transition to buprenorphine using the low dose initiation approach described above. A standard initiation approach should be avoided unless it has been more than 72 hours since last dose of no more than 30 mg of methadone.

Naltrexone

Naltrexone is an opioid receptor antagonist that blocks the euphoric and analgesic effects of opioids. The monthly extended-release injectable formulation can be used for opioid use disorder; however, it is not first-line therapy. In the general population, methadone and buprenorphine are associated with lower overdose rates compared to naltrexone.⁴⁸ Additionally, naltrexone initiation can be challenging because it requires patients to be abstinent from opioids for 7-10 days prior to initiation. Though there is no evidence of safety concerns of naltrexone in pregnancy, the evidence of efficacy is lacking compared to methadone and buprenorphine.^{49,50} In addition, for the pregnant population in which labor pain management is of concern, long-acting naltrexone, which lasts 6 weeks, may be difficult to safely stop far enough in advance. In these cases, patients should receive both neuraxial and regional anesthesia and nonopioid analgesics for adequate pain control.⁵¹

In a patient who is stable on naltrexone prior to pregnancy, naltrexone continuation can be considered on a case-by-case basis. Though not first-line therapy, naltrexone is better than no treatment for overdose prevention – newer research may support its initiation if patients are not interested in buprenorphine or methadone.⁵²

Management of Opioid Withdrawal

Management of opioid withdrawal is an important component of OUD treatment, whether during MOUD titration or for a patient who declines MOUD initiation. Withdrawal symptoms usually begin 6-12 hours after use of short-acting opioids (e.g., heroin, fentanyl), 24-36 hours after use of extended-release formulations (e.g., Oxycontin, MS Contin), and 24-72h after use of methadone.

Clinicians should ask patients about withdrawal symptoms and treat them aggressively. Opioid withdrawal is associated with high risk of return to use and patient directed discharge from the hospital, as well as low birthweight, preterm labor, and fetal stress.^{53,54} Opioid withdrawal can have variable onset and duration depending on if short- or long-acting opioids have been used. Withdrawal symptoms include restlessness/agitation, runny nose, tearing, myalgias, arthralgias, nausea, vomiting, diarrhea, and abdominal pain. Physical exam findings include pupillary dilation, yawning, sweating, runny nose, increased bowel sounds, and piloerection. The Clinical Opiate Withdrawal Scale can be used to quantify the severity of opioid withdrawal (Table 4).

Though opioid withdrawal or “detox” is not advised, guidelines recommend that methadone or buprenorphine can be used to stabilize acute opioid withdrawal, regardless of patient’s intent to continue as an outpatient.^{55,56,57} If patients prefer to be tapered prior to discharge, doses should be tapered slowly. For example, a patient started on methadone 40 mg daily could be tapered by 5-10 mg per day. Providers can also safely and legally use short-acting opioids to treat acute withdrawal in hospitalized patients.⁵⁸ For example, oxycodone 10 mg can be given every 4 hours for COWS > 8, with up titration as needed for adequate control of symptoms. If short acting opioids are started in the hospital settings to treat withdrawal, they must be tapered prior to discharge. Per DEA regulations, opioids cannot be prescribed for the sake of treating withdrawal in the outpatient setting.

Adjuvants for opioid withdrawal should be optimized to support patients as well. A list of adjuvants that are safe in pregnancy can be found in Table 5. Providers can check safety data for additional medications on Infant Risk Center.⁵⁹ or ReproTox⁶⁰

Managing Acute Pain

Pain control is beyond the scope of this article. Briefly, buprenorphine and methadone should be continued upon hospitalization and through birth.⁶¹ Splitting the total daily dose of buprenorphine and methadone into 3-4 times daily can optimize their analgesic effect, which is shorter than their effect on withdrawal symptoms.^{62,63} This split dosing can be continued through the postpartum period, if feasible. Long-acting naltrexone, on the other hand, should be stopped if opioid analgesics are required for labor or other acute indications.

MOUD alone does not provide adequate analgesia for acute pain. Therefore, a multimodal approach should be used, recognizing that higher dosages of opioids, if indicated, may be needed due to the patient’s opioid tolerance and hypersensitivity to pain.⁶³ A taper plan should be determined before discharge.

If further specifics on pain management are needed, we recommend the Consensus Statement on Pain Management for Pregnant Patients with Opioid-Use Disorder from the Society of Obstetric Anesthesia and Perinatology, Society for Maternal-Fetal Medicine, and American Society of Regional Anesthesia and Pain Medicine.⁵¹

Managing Overdose

Patients with clinically significant respiratory depression and other acute life-threatening opioid toxicity should be given naloxone. A retrospective cohort study found that pregnant persons are less likely to receive naloxone than non-pregnant persons.⁶⁴ Naloxone is safe in pregnancy and delay in administration can be life-threatening to both the pregnant person and fetus. The appropriate dose of naloxone is controversial. Therefore, the dosage should be titrated based on response to treatment; repeated dosages can be given. Buprenorphine overdose may require high dosages of naloxone (>2 mg) due to its high receptor affinity.⁶⁵ Given naloxone’s relatively short half-life, patients may require a continuous infusion to reverse an overdose. Both patient and fetus should be monitored after naloxone is administered.

Postpartum Care

OUD treatment and support should continue into the postpartum period. The postpartum period brings significant changes. Triggers for return to use can include loss of insurance and access to treatment, demands of childcare, sleep deprivation, and fear of losing custody of their children.¹³ Several studies have shown that risk of death secondary to overdose is highest in the first 2 years of the postpartum period.^{66,67} In fact, a recent study found that in pregnant persons with OUD and Medicaid insurance cumulative incidence of all-cause mortality within the first year postpartum was 6 times greater than the general population.⁶⁸ MOUD should be continued postpartum. A 2023 prospective cohort study found that postpartum use of MOUD was associated with 60% lower odds of opioid overdose death.⁶⁸

Breastfeeding

Breastfeeding facilitates both mother-infant bonding and infant immunity. Both methadone and buprenorphine are safe to use in breastfeeding, regardless of dose.⁶⁹ In fact, breastfeeding is associated with decreased severity of NOWS with less pharmacotherapy and shorter hospital stays for infants.⁷⁰ Recent guidelines state that patients who have discontinued non-prescribed substance use by delivery should also be supported in breastfeeding initiation.⁷¹ Those patients who have continued use and wish to breastfeed, can be supported in expressing milk to establish milk production. The decision as to when to start breastfeeding among this subset of patients should be a shared decision between the patient and clinicians of both mother and baby.

Child Welfare

Maternal substance use does not equate to inability to care for a child and should not automatically lead to child welfare involvement. The mother-infant dyad is foundational to the health and well-being of both individuals. Providers should collaborate with the infant's pediatricians, social workers, and the patient to create a plan that prioritizes family preservation with adequate psychosocial support services, including substance use disorder treatment.

Many pregnant persons who have opioid use disorder are concerned about court involvement and separation from their child. These fears may influence their treatment decisions.⁷² Mandatory reporting statutes are often misunderstood by providers and negatively impact the patient-clinician relationship, causing moral distress amongst health professionals. Providers caring for pregnant and postpartum individuals need to be aware of state laws and hospital policies on child welfare and substance use (see the If/When/How Prenatal Drug Exposure resource)¹⁷ and share this information with the patient so that they partner together.

Discharge Planning

Discharge planning should begin early on during admission. Options for ongoing OUD treatment after discharge range from outpatient care to residential treatment programs depending on the level of support needed and patient preferences. When possible, patients should be connected to programs and providers that have expertise in the care of pregnant and/or parenting patients with OUD and that provide dyad-focused care.

Patients who have been newly initiated on buprenorphine and methadone must have an identified prescriber or OTP, respectively, prior to discharge. Studies have shown discontinuation of MOUD increases overdose rates.^{73,74} All patients on buprenorphine should be provided with a prescription for buprenorphine until they can see their outpatient prescriber. Of note, as of 2023, all providers with current DEA registration that includes Schedule III authority can prescribe buprenorphine for opioid use disorder.⁷⁵ Similarly, patients on methadone should be given take-home dosages of methadone and established with an OTP. In 2022, the DEA announced that any DEA-registered hospital can dispense up to 3 days of methadone at one time.⁷⁶ All patients with OUD should be discharged with naloxone and taught how to administer it. For patients whose OTP is far from the hospital, providers can also advocate for methadone "guest dosing" or dispensing by a closer OTP if the newborn remains admitted for NOWS monitoring after parent discharge.

Social work and peer navigators should be engaged to help coordinate outpatient resources including postpartum psychosocial support services such as substance use disorder treatment and relapse prevention programs. A 2011 Cochrane review found that psychosocial intervention in addition to medications has been shown to improve retention in care and decreased use of opiate use compared to medication alone.⁷⁷

Discharge planning should be family-centered, considering partners, any previous children, and the newborn. A pregnant person is nearly five times more likely to use substances if her partner also uses⁷⁸; thus, engaging a patient's partner in treatment is crucial in effective care. As aforementioned, patients should also be screened for intimate partner violence and discharged to a safe environment. If the patient is postpartum, childcare and availability of parenting support should be considered. Childcare responsibilities are a stressor and can interfere with treatment engagement. There are few treatment centers with childcare available.⁷⁹

Conclusions

Pregnant persons with opioid use disorder face numerous barriers to care. The hospital setting offers a unique opportunity in which both general internists, family medicine clinicians, midwives, and obstetricians

can provide evidence-based, non-stigmatizing care to address their acute needs and connect them to long-term outpatient resources to improve both maternal and fetal outcomes. Ongoing clinical research is needed to optimize dosing strategies and evaluate newer formulations such as long-acting buprenorphine in hospital-based care.

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